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# Ten recommendations for improving the study of executive functions in sport and exercise

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**ABSTRACT**

Research on executive functions (EFs) has gained increasing attention in sport and exercise science and psychology. While EFs are widely discussed for their potential to enhance athletic performance, the direct evidence remains inconclusive due to methodological inconsistencies, conceptual ambiguity, and limited ecological validity in current research designs. This article addresses these challenges by offering ten recommendations and methodological guidelines for future EF research in sport and exercise contexts. It advocates for the use of standardized definitions, the development of ecologically valid and sport-relevant tasks, and the inclusion of diverse participant populations, while emphasizing the importance of multidisciplinary collaboration. The article also highlights key gaps, such as the weak causal evidence linking EF training to performance outcomes, and the variability of such effects across populations. By integrating evidence-based guidance with theoretical insights, the proposed recommendations aim to support rigorous, interdisciplinary, and applied EF research with greater translational value for sport and exercise psychology.

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Research on executive functions (EFs), a set of three core cognitive processes including inhibitory control, working memory, and cognitive flexibility, has gained significant traction in sport and exercise science and psychology in recent years (see Furley et al., 2023). These functions are considered essential for successful performance across various domains, from academic achievements to complex motor tasks (Diamond, 2013). In the context of sport and exercise, EFs have been proposed as significant contributors to athletes' ability to plan, adapt, and make rapid decisions under pressure. At the same time, there is growing interest in the potential for physical activity and sports participation to enhance EF, leading to discussions about the mechanisms underlying this relationship. However, research in this area continues to be hindered by a range of methodological and conceptual limitations (Furley et al., 2023; Wood et al., 2023). Studies in this field often suffer from inconsistent definitions of EF (Eslinger, 1996; Karr et al., 2018), limited

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validity and reliability of assessment tools (Diamond, 2020; Nyongesa et al., 2019), and poor standardization in measurement techniques, study design, and data collection procedures (Furley et al., 2016; Packwood et al., 2011). These issues have contributed to an ambiguous evidence base, making it difficult to draw conclusions about the relationship between EF and athletic performance, or about the efficacy of EF training interventions (Furley et al., 2023; Sala & Gobet, 2019).

This lack of robust empirical evidence has significant consequences for practice. Claims that EF assessments can predict talent or that EF training directly enhances performance often exceed the supporting science, fuelling commercially driven programmes marketed as 'scientifically proven' despite weak evidence (Bailey et al., 2018; Harris et al., 2018). Such premature applications risk undermining the credibility of sport and exercise psychology and highlight the need for greater methodological rigour.

In response, this article sets out ten recommendations for academics, designed to enhance the quality and precision of EF research. These recommendations span methodological foundations, contextual influences, and population-level considerations, offering both immediate guidance for improving current practices and long-term priorities for shaping the field's future direction. Specifically, we emphasize the importance of establishing clear and unified definitions of EFs, adopting standardized and ecologically valid measurement tools, and improving study design through longitudinal approaches and greater transparency. We also highlight the value of considering developmental factors, psychological moderators, and participant diversity, alongside fostering interdisciplinary collaboration to ensure inclusivity and a more holistic perspective. Collectively, these recommendations aim to provide researchers with a practical framework for generating more robust, reliable, and meaningful insights into the relationship between sport, physical activity, and EFs.

## Recommendation 1: Promote Theory-Driven and Evidence-Based Definitions of Executive Functions

> Definitions of EF should be grounded in solid theoretical approaches making sure to distinguish between tasks and theoretical concepts.

The term EF first appeared in the literature in the early 1980s (Lezak, 1982), building on earlier work that had used the term 'executive' to describe frontal lobe functioning (Pribram, 1973) and mental control processes (Baddeley & Hitch, 1974). EFs are assumed to be central in understanding human thought and behaviour. They are typically described as a family of cognitive processes that engage when behaviour must be adjusted to novel, unanticipated circumstances and humans cannot rely on their automatic, instinctive or intuitive response tendencies. A further central tenet of this construct is the assumption that EFs are capacity-limited, which helps explain why individual differences in EF measures have been reported to be strongly correlated with various valued achievements and life outcomes, such as school grades, academic success, and even the quality of social relationships (Diamond, 2013; Katz et al., 2018).

Despite the cross-disciplinary interest in EF, the conceptualization of EF presents notable challenges in the field due to the complex and multifaceted nature of the construct (Miyake et al., 2000; Yangüez et al., 2023). With at least 33 different definitions and 68 subcomponents identified within the EF literature, this significantly complicates cross-study comparisons and the creation of a unified EF model (Eslinger, 1996; Karr et al., 2018; Packwood et al., 2011). Among the various challenges that hinder the development of a coherent theoretical framework, one major issue is the lack of consensus among researchers regarding the core dimensions, definitions, and measurements of EF. Some researchers conceptualize EF as a unified structure (Miyake et al., 2000), while others regard it as a collection of independent abilities (Eslinger, 1996; Karr et al., 2018).

One of the most influential models on EF (Diamond, 2013) distinguishes three core EFs from one another: Working memory (WM), inhibitory control, and cognitive flexibility (Diamond, 2013). Working memory can be defined as the cognitive mechanisms capable of retaining a small amount of information in an active state for use in ongoing tasks (Baddeley & Hitch, 1974; Miyake & Shah, 1999). Inhibitory control involves self-regulation by controlling one's attention, thoughts, and emotions. Its importance has been illustrated by Walter Mischel's seminal work on delay of gratification (Mischel et al., 1989). Typically, this involves overriding impulses resulting from internal or external stimulation and instead adjusting behaviour in an intentional, goal-directed manner (Simpson et al., 2012; Watson & Bell, 2013). Cognitive flexibility refers to the ability to flexibly switch between different mindsets or tasks, taking a different perspective on matters, and quickly and flexibly adjusting to new information and changing circumstances (Diamond, 2013). These core EFs are assumed to comprise the core building blocks of higher-order EFs such as reasoning, problem solving, and planning (Collins & Koechlin, 2012), which together underpin fluid intelligence. Yet the empirical evidence supporting this hierarchical structure remains limited.

The unity/diversity framework proposed by Miyake et al. (2000) includes three core functions – updating, shifting, and inhibition – that, while not clearly distinguishable, are intercorrelated (see Panesi et al., 2022 for a current discussion on the distinction between working memory capacity and updating). The original three-factor model has since been modified to a nested-factor model, where a single common EF factor encapsulates the shared variance among updating, shifting, and inhibition (Friedman & Miyake, 2017; Miyake & Friedman, 2012). In applied samples such as athletes, evidence generally supports the nested structure, with the unity/diversity distinctions also detectable to a lesser extent (Brimmell et al., 2025). Although the unity/diversity framework has been widely replicated and is among the most prominent models, confirmatory factor analyses have revealed that EF measurement models frequently fail to achieve adequate fit across studies for all age groups (Karr et al., 2018; Yangüez et al., 2023; Sambol et al., 2023). This inconsistency raises concerns about model stability and replicability. The lack of consensus on the conceptualization of EF complicates theoretical clarity and comparability across studies. To address this, efforts should focus on integrating existing frameworks, such as the unity/diversity model (Friedman & Miyake, 2017; Miyake et al., 2000). Standardizing definitions and subcomponents of EF is crucial to enhance theoretical consistency and enable meaningful comparisons across studies. Without this standardization, findings remain fragmented, limiting their reproducibility and practical application.

However, reaching theoretical consensus has been shown to be especially challenging in the field of EFs as this is closely connected to the difficulty of establishing a clear and unified definition of EF. A further problem in this regard (which also touches on our second recommendation: measurement) stems from a close link between definition and measurement of EFs. This problem has often been described in the broader field of cognitive measurement (Mashburn et al., 2024) as it invites circular definitions like 'intelligence is what intelligence tests measure' or 'EFs are whatever EF tests measure'. Regarding the most contemporary theoretical definitions, we recommend Mashburn et al.'s (2024) reviews of the 'unity/diversity theory', 'maintenance and disengagement theory', and 'process overlap theory'. Maintenance and disengagement theory provides a more granular look at how updating actually works by positing that the updating process isn't a single action, but a two-step process of keeping the right things in and kicking the wrong things out (e.g. Mashburn et al., 2020). Process overlap theory (Kovács & Conway, 2016) was set out to explain why different cognitive tests – like math, verbal, and spatial reasoning – all tend to correlate with one another, with the underlying correlation becoming known as g, the general factor of intelligence. While a detailed synthesis of these theoretical approaches is beyond the scope of this review, our central point here is to ground definitions of EF in solid theoretical approaches (e.g. Mashburn et al., 2024) and make sure to distinguish between tasks and theoretical concepts. A way forward in this regard was proposed by Gigerenzer (2017), in which the challenge of maintaining theoretical hygiene is addressed by advocating for the integration, or at minimum the convergence, of extant theoretical frameworks – i.e. 'unity/diversity theory', 'maintenance and disengagement theory', and 'process overlap theory'. Moving forward, researchers should explicitly link EF tasks to theoretical constructs to enable coherent and comparable findings across studies.

## Recommendation 2: Employ Standardized and Validated Measurement Tools

> To improve measurement quality in EF research, researchers should use validated test batteries that include multiple tasks per construct and report reliability transparently.

A central challenge for both EF measurement and interpretation in sport and exercise research is task impurity, the tendency of tasks to recruit multiple cognitive processes simultaneously, making it difficult to isolate specific EF components. Latent-variable modelling or composite scoring represents the most reliable strategy for separating genuine construct variance from task-specific noise (Friedman & Miyake, 2017; Miyake et al., 2000; Yangüez et al., 2023). However, at the single-task level, apparent 'core' EF tasks (i.e. single-task indices) may partly capture overlapping task-specific skills rather than distinct cognitive constructs. This is precisely why single tasks should be treated as insufficient indicators and researchers should rely on latent-variable or multi-task approaches (Brimmell et al., 2025; Miyake & Friedman, 2012; Sambol et al., 2023). Although complete isolation of EF components is rarely feasible, or even desirable, in applied sport and exercise contexts, combining multiple complementary tasks per construct and applying composite or latent-variable approaches can substantially strengthen construct validity while preserving ecological relevance. Such approaches, however, typically require larger sample sizes and sufficient trial numbers, which constrains their applicability in many sport and exercise studies and should therefore be recognized as an important methodological limitation (Brimmell et al., 2022; Yangüez et al., 2023). This issue lies at the heart of ongoing debates on how to measure EFs and determine the components a task truly requires. Furthermore, many tests suffer from extraneous demands, procedural overlap, and variability in task design, measurement techniques, and contextual factors, all of which introduce task-specific variance and lead to inconsistent results (Friedman & Miyake, 2017). It is also important to note that the proliferation of outcome variables in EF studies can reflect a tendency to test multiple measures until a statistically significant result is found, which may bias findings (Dwan et al., 2008).

*Inhibitory control* is commonly assessed with tasks such as the Stroop, Simon, Flanker, anti-saccade, go/no-go, and stop-signal paradigms, which typically contrast congruent (primarily nonexecutive) with incongruent (nonexecutive plus executive) trials. Because these measures reflect a mixture of processes, using simple contrast scores can lead to contamination. Recommended practices include employing multiple tasks, reporting psychometric indices (e.g. test–retest reliability, ICCs), and, where feasible, using model-based indices such as drift-diffusion parameters[1] to extract more stable executive signals (Sambol et al., 2023; Yangüez et al., 2023). Another challenge is the heterogeneity of indicators: some studies emphasize reaction times, others accuracy, complicating cross-study comparisons. While accuracy-based measures often yield higher reliability, they are also vulnerable to ceiling and floor effects. Model-derived parameters such as drift rate frequently demonstrate stronger psychometric properties, provided sufficient trial numbers and model convergence are achieved. Regardless of indicator choice, studies should transparently report reliability estimates and preprocessing criteria (Yangüez et al., 2023). Tasks like the Victoria Stroop, Contingency Naming Test, and Stop-Signal Task combine speed and accuracy but emphasize accuracy for robust results (Sambol et al., 2023). Rather than relying on raw difference scores (e.g. congruent–incongruent), which often demonstrate poor reliability, researchers should consider alternative strategies such as residualized scores, latent-variable approaches, or model-based parameters. If difference scores are used, their psychometric properties (e.g. ICCs, test–retest estimates) should be reported to ensure interpretability in individual-differences research (Westfall & Yarkoni, 2016; Yangüez et al., 2023).

*Cognitive flexibility* is typically assessed using task-switching paradigms, but these tasks also recruit processes such as processing speed, inhibition, and interference control (Sambol et al., 2023; Yu et al., 2017). Further complicating comparability, studies often operationalize cognitive flexibility using different performance metrics, which can limit comparability across studies (Yangüez et al., 2023). Researchers should therefore use standardized and transparently reported performance metrics and, where possible, complement task-switching paradigms with additional measures of cognitive flexibility rather than relying on a single task or performance metric. Where data permit, model-based parameters (e.g. drift-diffusion estimates) may enhance convergence across paradigms, though transparent reporting of model fit and reliability remains essential (Yangüez et al., 2023).

*Working memory* (WM) is commonly assessed with complex span tasks (e.g. counting span, operation span), N-back tasks, digit-span, and Corsi Block tests. However, these often engage additional processes such as inhibition, cognitive flexibility, attention, or processing speed (Diamond, 2013; Lezak, 1982), reducing their specificity. The N-back's variable cognitive load can also confound results. WM assessment can be improved by controlling load variability, combining complementary tasks (e.g. digit span with pure visual–spatial WM measures), and using standardized computerized batteries, such as the AWMA or CANTAB (Alloway et al., 2008; Luciana & Nelson, 2002). When complete standardized batteries are not available for a given target group, researchers should include at least two complementary tasks per EF component to improve construct coverage and support latent-variable modeling, while considering participant burden.

In sum, employing standardized and validated test batteries, while acknowledging the interdependence of EF components, will improve both accuracy and comparability in sport and exercise related EF research. Beyond psychometric soundness, tasks should be selected for ecological validity, aligning closely with the cognitive demands of competitive sport. Importantly, transparent reporting of reliability indices, trial numbers, exclusion criteria, and model diagnostics should become standard practice to ensure reproducibility and applicability (Brimmell et al., 2022; Yangüez et al., 2023).

In practice, researchers should prioritize multi-task, standardized, and psychometrically validated measures, alongside transparent reporting, to ensure EF assessments meaningfully reflect the constructs of interest.

## Recommendation 3: Employ More Ecologically Valid Tasks

> To improve the real-world relevance and transferability of EF assessments in sport and exercise, researchers should prioritize ecologically valid tasks that reflect the cognitive demands athletes face in their sporting environment. At the same time, care must be taken to maintain sufficient construct validity by avoiding tasks that conflate multiple EFs.

A good task of EF for athletes should broadly facilitate two things. First, the task should allow us to make inferences about a single specific EF (i.e. avoid impurity issues mentioned above). Second, the demands placed on the individual in the task should require them to utilize the target EF in the same way their sport would (i.e. tasks should be ecologically valid). Tasks of EF used with athletes often lack this ecological validity due to the use of artificial constraints or laboratory settings which are void of sport-specific context (Brimmell et al., 2022). For example, Brimmell et al. (2025) used a laboratory-based Stop Signal Task to measure inhibitory control in a sample of athletes. Although the task was deemed a reliable measure of inhibitory control (α = .88 reported in Brimmell et al., 2025), the task involved fine motor movements (i.e. a finger press on a keyboard) in response to the presentation of contextless letter stimuli. The task, therefore, was not particularly representative of sport or the movement patterns required within most gross motor sports (i.e. low in ecological validity). This might help explain why far-transfer after EF training is limited (Furley et al., 2023).

Increased ecological validity should not come at the detriment of precise EF measurement. Therefore, we recommend that EF researchers focus on finding a middle ground between the currently used EF tasks (e.g. Stop Signal Task; Verbruggen et al., 2019), which are designed to target a specific EF but lack sport- and exercise-specific context (i.e. lack ecological validity), and sport- and exercise-specific tasks that require representative movement patterns, stimuli processing, and environments (Brimmell et al., 2022). In doing so, we may be more able to capture the dynamic and complex nature of sporting environments, and as Green et al. (2019) suggest, enhance the efficacy and applicability of EF assessments in sport and exercise psychology. Even though the middle ground may be ideal– and our recommendation – in moments where isolation of an EF is key, measurement precision is more important; and when performance is key, ecological validity has to be a key factor. For a practical framework to support this, we can look at the debate between opposed and unopposed practice in sport and exercise coaching. Unopposed practice is dominant and involves athletes practicing motor skill in isolation (Parry et al., 2025). However, opposed practice (with roots in ecological dynamics; Roca & Ford, 2020) suggests that context is key, and that action never occurs in a vacuum (Parry et al., 2025). Here, we suggest that we stop looking to examine EFs in isolation (i.e. unopposed practice) and instead design EF tasks that factor in more ecologically valid opposed components.

Historically, domain-general EF tasks (i.e. tasks with no contextual information relevant to a population or environment) have been used to assess deficits in cognitive control, specifically pertaining to the prefrontal cortex, within clinical populations or individuals with brain lesions (Friedman & Robbins, 2022). As a result, it is no surprise that such tasks fail to capture the complexities of competitive sport performance, given that this was not their intended purpose. This highlights an important issue when transferring such tasks across domains, a typical approach in sport and exercise science and psychology (Furley et al., 2023; Finkenzeller et al., 2023). To alleviate this issue and boost ecological validity for sport and exercise, we recommend that individuals build new EF tasks that begin in the sporting and exercise world. That is, look for naturally occurring scenarios within matches or events where a particular EF might be important.

For example, in soccer, a teammate may present as a viable passing option until an opponent steps into the passing line. In this moment, the ball-carrier needs to inhibit the planned movement of a pass and instead select another option (e.g. hold the ball, pass elsewhere, dribble forward). Such moments reflect quite naturally the core process of a Stop Signal Task, where certain stimuli are associated with completing an action (i.e. 'go' trials), while others require inhibition (i.e. 'stop' trials). To replicate this paradigm in the soccer context, we could have a viable passing option presented virtually (either via computer or a virtual reality headset) before having an opposing virtual player move to 'mark' the passing option, making them no longer viable. Here, individuals would have to learn to withhold the passing response (to inhibit) to avoid giving away the ball.

In their recent meta-analytic review, Kalén et al. (2021) outlined a key conceptual split between cognitive skills and cognitive functions in athletes (terms similar to domain-general and sport- and exercise-specific used in the present work). Cognitive skills are measured through tasks that capture key elements of the specific sport or exercise (e.g. highly relevant stimuli or response types) to provide higher ecological validity (Araujo et al., 2007). Alternatively, cognitive function tasks lack sport and exercise-specific context but are more established and rigorous methods of measuring the target EF. As such, we recommend that researchers select an EF task that is suitable for their current needs. That is, perhaps domain-general tasks fit to their purpose (i.e. clinical, or post-concussion, assessment), but their purpose does not extend to the unique environment of sport and exercise. Here, Kalén et al. (2021) provide researchers and practitioners with the appropriate language, and a clear justification, for two lanes of EF measurement (i.e. a domain-general and sport- and exercise-specific split).

It is positive that such a review exists and allows us to comment on the use of sport and exercise-specific EF tasks; however, there is still a dominance of domain-general EF tasks being utilized. The impact of this is that sport and exercise researchers or practitioners are placed into a situation of uncertainty due to a confusion about which type of EF training task would best serve their purpose (i.e. cognitive skills or function tasks). Evidence from a recent meta-analytic review suggests that higher-skilled athletes outperform lesser-skilled athletes on both measures of cognitive skill and function (Kalén et al., 2021). However, the size of the effect was larger for sport and exercise-specific (i.e. cognitive skill) tasks compared to domain-general (i.e. cognitive function) tasks. As such, these results support our main recommendation for the practical implementation of EF training, which is to prioritise sport- and exercise-specificity where ecological validity is critical, while still considering domain-general tasks when precise measurement of isolated executive functions is needed.

Consequently, future EF tasks in sport and exercise should integrate real-world contexts while preserving the ability to isolate target executive functions.

## Recommendation 4: Incorporate Longitudinal Designs

> To better understand how EF develop and change in response to sport and physical activity, researchers should incorporate longitudinal designs that track athletes over time. Such designs are essential for identifying developmental trajectories, critical periods for intervention, and the long-term impact of training or competition on EF.

Longitudinal designs provide valuable insights into how EFs evolve with age (though age is an important factor to consider in all EF work – see Recommendation 9), physical activity, or targeted interventions. This is largely because the human brain is not 'fixed' (i.e. neuroplasticity is possible), with changes in neuronal count, synaptic connections, and neuron myelination being just some of the structural changes that occur throughout the lifespan (O'Hare & Sowell, 2008). Regular sport and exercise activity is a large contributor to this plasticity and has been associated with the promotion of brain-derived neurotrophic factor (i.e. a key factor in the creation of new neurons; Zhou et al., 2022). Therefore, any maintained sport and exercise participation will promote neuronal growth, alter brain structures, and thus can impact an individual's EF. This tendency for change makes applying the findings of any cross-sectional work beyond the short-term very difficult and instead really emphasises the need for longitudinal work and understanding growth trajectories and inter-individual differences.

In addition, the trajectory of neuronal changes does not always adhere to the typical inverted-U model (i.e. steady increases from birth to early adulthood, peaks and then gradually declines into older age; O'Hare & Sowell, 2008) and participating in sport and physical activity can play a role in positive neuronal changes at various stages of development (Zhou et al., 2022). This presents a unique opportunity to not only follow EF development from early years to adulthood, but also from adulthood to older adulthood. Therefore, the next recommendation is to conduct longitudinal work across the lifespan at the macro- (i.e. from infancy to late adulthood) and micro-level (i.e. within a single period of development [e.g. infancy]).

The prefrontal cortex is a key neural area for EF and is considered to be one of the regions most capable of adaptation and the last region to fully develop (Waxer & Morton, 2011). Given its high adaptability, late development, and relevance to EF, research outside of sport and exercise (e.g. Diamond, 2013; 2016) has suggested that targeting the prefrontal cortex through EF training during early years is key. Indeed, Diamond (2016) noted that children of young ages can differ in EF ability at a single point in time and such differences can persist or increase in adulthood. However, whether this is a critical period for young people engaging in regular sport and exercise remains unknown. While it seems logical to assume so, this relationship is complicated by the naturally higher levels of physical activity during this stage of life. As of right now, we do not know whether exercise can enhance initially low levels of EF and/or further enhance naturally high EF in young individuals. Therefore, our recommendation to use longitudinal designs should also allow us to understand if there are critical periods within development where EF training and/or assessment might be most useful (e.g. evidenced by large increases in EF during a particular period). If we have this information, we can devise more effective and targeted interventions and ensure resources are used at the right moment.

The sporting industry (e.g. soccer academies) is always looking for optimal ways to spot and/or nurture talent. The idea of EF assessment rose in popularity as a new and effective way for talent identification. However, this avenue has, so far, been less fruitful than originally advertised or hypothesized (or at least, less empirically supported; Furley et al., 2023; Harris et al., 2018). While initially damning, the mixed results regarding the impact of EF assessment may be due to the current plethora of cross-sectional or short-term utilization of EF measurement and/or training, and the limited amount of work on the long-term effect of EF training upon sport performance (Harris et al., 2018). The lack of longitudinal work likely stems from the difficulty of such work. That is, there are many costs (e.g. time, effort, and money) that act as barriers to conducting this work. In addition, it is rare that researchers in this area will have access to the same individuals for an extended period of time and thus, cannot conduct longitudinal work. Therefore, our final recommendation here is for academics to work with sporting clubs and organizations, for example soccer academies, that have individuals stay with them for an extended period of time. Doing so will allow us to conduct this kind of research that is vital for the success of understanding the EF-sport/exercise relationship.

## Recommendation 5: Collaborate with Multidisciplinary Teams

> There is a need for multidisciplinary collaboration, bringing together expertise from exercise science, psychology, neuroscience, and sports medicine. Such partnerships enhance methodological rigor, enable innovative designs, and ensure that EF interventions are both scientifically robust and practically relevant.

Advancing EF research in sport and exercise science requires genuine collaboration across disciplines. Bringing together experts from exercise science, psychology, neuroscience, and sports medicine allows researchers to address the complexity of EF from multiple angles (Green et al., 2019). For example, exercise scientists provide insights into training protocols, psychologists contribute expertise on EF assessment and mechanisms, and neuroscientists explore brain changes associated with exercise.

Multidisciplinary teams not only enhance the methodological rigour of studies but also facilitate innovative research designs – such as combining behavioural tasks with EEG, fMRI, or heart-rate variability measures – to capture the full scope of how physical activity influences cognitive function (Stillman et al., 2016). These collaborations also improve translational impact by ensuring that interventions are both scientifically grounded and practically relevant to target populations (e.g. youth athletes, older adults, or clinical groups).

In sum, tackling the multifaceted challenges in EF research demands interdisciplinary thinking. This approach can produce more robust, ecologically valid findings and ensure that interventions in sport and exercise are both effective and evidence-based.

## Recommendation 6: Control for Confounding Variables like Fitness Levels and Physical Fatigue

To enhance the reliability and interpretability of EF research in sport and exercise, researchers should systematically identify and control key confounding variables. These may include physiological factors that are particularly salient in sport and exercise contexts (e.g. fitness levels and physical fatigue), as well as broader individual and contextual influences such as age, pubertal status, baseline information-processing speed, socioeconomic context, and family sport involvement. If unaccounted for, such factors can introduce unwanted variability and lead to misleading inferences about EF mechanisms or intervention effects.

Fitness and physical fatigue represent especially important confounds in sport and exercise research because both directly influence EF performance and can vary substantially within and between participant groups. Physical fatigue and fitness modulate core EF components such as attention, cognitive flexibility, and inhibitory control (Best & Miller, 2010; Yangüez et al., 2023). For instance, tasks assessing cognitive flexibility may yield markedly different results depending on whether participants are tested in a rested state or following physical exertion (Lambourne & Tomporowski, 2010). Similarly, differences in baseline fitness can shape cognitive responses to training or acute exercise, thereby confounding comparisons across individuals or groups.

Supporting this, Huijgen et al. (2015) demonstrated that elite youth soccer players, who engaged in significantly higher physical training hours, also scored higher on composite EF tasks – including inhibition, shifting, and updating – compared to sub-elite peers. Notably, when physical training hours were statistically controlled as a covariate, the differences in EF between groups were reduced, though still significant, indicating that training volume partly drives EF disparities. This finding underscores the importance of including physical activity or training hours as covariates in EF research to avoid attributing cognitive differences solely to other variables.

To address these challenges, researchers should employ systematic assessments of fitness or physical activity using standardized measures, such as maximal oxygen uptake ($VO_2$ max), lactate threshold, or validated field protocols (e.g. the Yo-Yo intermittent recovery test; Bangsbo et al., 2008), or self-reported physical activity questionnaires (e.g. Brimmell et al., 2021). These objective benchmarks facilitate participant stratification or matching in experimental designs, helping isolate the effects of cognitive manipulations from those related to fitness variability. Controlling exertion during EF testing is equally crucial, with standardized protocols (e.g. Wingate test) and objective monitoring tools (heart rate monitors, power metres), alongside subjective scales like the Borg Rating of Perceived Exertion (Borg, 1982), enabling precise control of fatigue levels and reducing outcome variability. Additionally, environmental temperature can modulate the effects of exercise on cognitive performance, with moderate to vigorous exercise particularly sensitive to cold or heat, and executive function tasks appearing most affected (Donnan et al., 2021).

Rest and recovery periods should also be standardized based on prior physical activity intensity and duration to minimize residual fatigue effects on EF assessments (Brisswalter et al., 2002). Additionally, external factors such as circadian rhythms, sleep quality, hydration, and environmental conditions (temperature, altitude) can influence fatigue and cognition and should be controlled by scheduling consistent testing times and monitoring participant sleep and hydration where possible.

While controlling for fitness, fatigue, and related physiological and environmental factors is essential for improving the reliability of EF research in sport and exercise, such controls must be applied judiciously. The choice and extent of control should be guided by the study's aims and design – whether isolating cognitive processes under tightly standardized conditions or capturing ecologically valid performance in real-world scenarios. Importantly, fitness and fatigue are not exhaustive sources of confounding in EF research; researchers should pre-specify and justify covariates based on theory, participant characteristics (e.g. developmental stage), and contextual influences. Tailoring these controls to the research question allows investigators to balance internal validity with practical relevance, ensuring findings that are scientifically robust and contextually meaningful.

## Recommendation 7: Distinguish Between Different Types of Exercise

> To enhance the specificity and effectiveness of EF interventions, researchers should systematically investigate how different exercise modalities – such as aerobic, resistance, and combined training – affect distinct EF domains. Understanding these differential effects is key to developing targeted and evidence-based cognitive enhancement strategies through physical activity.

Different exercise types engage unique physiological pathways that may selectively influence EF components. Aerobic exercise has been linked to improvements in cognitive flexibility and working memory, likely through mechanisms such as increased cerebral blood flow, neurogenesis, and the upregulation of brain-derived neurotrophic factor (Liu et al., 2020). Resistance training, by contrast, appears to enhance inhibitory control and cognitive flexibility, potentially via hormonal responses and structural brain changes, including increased grey matter volume and improved white matter integrity (Herold et al., 2019; Wilke et al., 2019). Emerging evidence suggests that combining aerobic and resistance training may yield additive or synergistic benefits by leveraging the distinct strengths of each modality (Zhang et al., 2024).

In addition to these broad categories, the cognitive demands inherent in specific sports and exercise contexts should be considered. Team sports, such as football or basketball, require rapid decision-making and cognitive flexibility in dynamic, unpredictable environments (Vestberg et al., 2017), whereas individual sports, like golf or archery, demand sustained attention and inhibitory control for optimal performance (Voss et al., 2010). Studying how these distinct activity contexts influence EF components can help design interventions that align with the cognitive requirements of particular sports, enhancing both performance and training relevance.

Methodologically, isolating modality-specific effects requires careful experimental control. Researchers should compare standardized aerobic, resistance, and combined protocols while systematically manipulating variables such as intensity, duration, and frequency. For example, high-intensity interval training (HIIT) may produce acute neurochemical changes – including elevated dopamine and norepinephrine – that enhance attention and processing speed (McMorris et al., 2016), whereas moderate-intensity continuous exercise may promote longer-term adaptations beneficial for working memory and cognitive flexibility (Stillman et al., 2016). Participant characteristics, such as age, baseline fitness, and initial cognitive status, should be accounted for, as these can moderate the relationship between exercise type and cognitive outcomes (see Recommendation 8). Tailoring protocols to these factors can further enhance the precision and efficacy of EF interventions.

Differentiating between exercise modalities is not merely descriptive but a strategic methodological decision that should align with the research question, target population, and intended application. By integrating physiological mechanisms, sport-specific cognitive demands, and rigorously controlled protocol variables, future research can produce findings that are both mechanistically informative and practically actionable. Such work will advance evidence-based training programmes capable of optimizing both cognitive and physical performance, with implications for athletes, older adults, and anyone seeking to improve cognitive health through physical activity. This approach also underscores the value of interdisciplinary collaboration between exercise science, neuroscience, and psychology (see Recommendation 5) to fully understand and leverage the relationship between physical activity and EF.

## Recommendation 8: Consider Psychological Moderators and Mediators

> Future research should systematically explore how psychological factors such as motivation, anxiety, and self-efficacy, moderate or mediate the relationship between exercise and EFs. Understanding these individual differences is critical for tailoring interventions and maximizing cognitive benefits across diverse populations.

Psychological factors such as motivation, anxiety, and self-efficacy significantly influence how exercise affects EFs (Bailey et al., 2018), often acting as moderators or mediators in the relationship between physical activity and cognitive performance. The well-documented associations between EFs and important life outcomes (Katz et al., 2018) have driven interest in cognitive training to enhance performance across domains (Rohde & Thompson, 2007; Stieff & Uttal, 2015; Wright et al., 2008).

Findings from neuroscience suggest EFs are malleable, particularly during childhood and adolescence (Casey et al., 2005), which has led to the commercial rise of cognitive training programmes claiming to 'train your child's brain' or 'boost working memory' (Serpell & Esposito, 2016). However, the effectiveness of such programmes remains highly debated (Boot et al., 2013; Melby-Lervag & Hulme, 2013; Shipstead et al., 2012; Simons et al., 2016). Psychological factors may partly explain why some individuals benefit more than others, raising the question of whether these interventions benefit all individuals equally or only specific groups.

One recurring claim in the literature is that individuals with lower EF, such as children with learning disabilities (Franceschini et al., 2013; Klingberg et al., 2005), people with schizophrenia (Biagianti & Vinogradov, 2013), traumatic brain injury (Hallock et al., 2016), or older adults (Anguera et al., 2013; Whitlock et al., 2012), may benefit most from training. Yet, findings are inconsistent. Some studies support greater gains among low-performing individuals (Jaeggi et al., 2008; Zinke et al., 2014), while others report greater benefits for those with higher baseline EFs (Bürki et al., 2014; Foster et al., 2017; Guye et al., 2017), or no relationship at all (Simons et al., 2016).

These mixed outcomes highlight the role of individual differences such as age, personality, or baseline EF levels in moderating training effects (Chein & Morrison, 2010; Shah et al., 2012; Zinke et al., 2011; Studer-Luethi et al., 2012). Unfortunately, our understanding of which interventions work best for which populations remains limited. It is unlikely that a single training protocol will generalize across groups. For instance, cognitive exercises for individuals with schizophrenia may differ substantially from those designed for healthy adults or children with attentional deficit hyperactivity disorder (Green et al., 2019).

EF training goals also vary: restoring lost function (e.g. in aging adults), improving atypical function (e.g. in developmental disorders), or enhancing typical function (e.g. in healthy young adults). While grouped under one umbrella, these interventions can differ markedly in design and expected outcomes. Moreover, factors such as belief in the intervention, effort, and persistence may influence results (Anguera et al., 2012; Jaeggi et al., 2011; Moreau, 2022). However, such claims have also been criticized for placing blame on participants when benefits are not observed (Moreau, 2022).

Finally, contextual moderators like stress (Liston et al., 2009), sleep deprivation (Nilsson et al., 2005), and physical fitness (Best & Miller, 2010) may also affect outcomes. Studying such effects requires large sample sizes, yet underpowered studies remain a challenge in EF research (Westfall & Yarkoni, 2016).

## Recommendation 9: Examine the Role of Age and Developmental Stages

> To design effective and developmentally appropriate EF interventions, researchers should systematically account for age and developmental stage when assessing or training EFs. EF development is gradual and influenced by neurobiological, environmental, and experiential factors, making age a crucial variable in both research and applied settings.

EFs differ in structure across different age groups as they are assumed to develop gradually from infancy to early adulthood. The development of these functions is deeply rooted in brain maturation, particularly in the prefrontal cortex, and is shaped by genetic, environmental, and experiential factors (e.g. Diamond, 2013). Key publications (Anderson, 2002; Best & Miller, 2010; Zelazo & Carlson, 2012) on the development and maturation of EFs can be summarized as assuming the following different stages in development – albeit it is important to note that these developmental stages are oversimplified and individual differences need to be recognized. Nevertheless, we consider these developmental stages to have important heuristic value for conducting research and applied practice in the area of EF.

### *Infancy*

Starting from 0–2 years, the emergence of attention and early self-control becomes apparent. Coinciding with prefrontal cortex maturation, early forms of self-regulation and delayed gratification like waiting for several seconds, can be observed in babies and young toddlers. More specifically, simple attention control (observing and focusing on objects), early inhibition (not reaching for forbidden objects), and short-term memory (finding a toy they had just been playing with) become evident in an infant's behaviour (Best & Miller, 2010).

### *Toddlerhood*

In the range from 2–3 years, inhibition, (working) memory, and basic rule-following improve. Typically, kids begin to follow rules better, they become able to shift attention between tasks, and suppress behavioural impulses in simple tasks (e.g. 'Simon Says' game). These changes are assumed to develop hand-in-hand with the onset of language development, which plays an important role in developing strategies for self-control (Best & Miller, 2010).

### *Pre-school*

From 3 to 5, typically rapid gains in core EFs occur and cognitive flexibility begins to emerge. Often this phase is assumed (Best & Miller, 2010; Zelazo & Carlson, 2012) to be the major developmental period of EFs. Behavioural observations, like waiting turns on the playground, resisting immediate temptations, following multi-step instruction, or differential weighting of rules signal strong improvements in inhibition, working memory and beginning cognitive flexibility.

### *Early school age*

In the range from 6–9 years, children show more planning and goal-directed behaviour, and kids are able to use EF in school tasks like math, reading comprehension, and general problem solving. Diamond (2013) assumes these behavioural observations to coincide with neuroscientific evidence demonstrating increased connectivity in prefrontal circuits.

### *Late childhood to beginning adolescence*

From about 10–14 years, more abstract reasoning, monitoring and reflecting on one's own behaviour and emotions, and strategic thinking (i.e. meta-cognition) become apparent in a child's behaviour. These abilities enable them – amongst other things – to take different perspectives, manage conflicting rules, and take long-term consequences of their momentary behaviour into consideration. In this phase, some studies have also suggested that in particular, cognitive flexibility and strategic thinking show the largest improvements (Diamond, 2013; for a review).

### *Late adolescence to early adulthood*

In this longest (approximately 15–25 year) and probably most significant EF developmental period, peak EF functioning is assumed to occur as a consequence of full prefrontal cortex maturation. Numerous scholars argue that full EF maturation occurs as late as 25 years (Anderson, 2002; Diamond, 2013) and serve their full potential in complex, long-term goal pursuit, self-reflection, emotion regulation, risk assessment, and adaptive behaviour in dynamic environments.

### *Consequences of the prolonged development of EFs for sport and exercise research*

These developmental differences imply that interventions should be tailored to the developmental phase of the individuals involved. For example, younger athletes may benefit from holistic core EF training that targets integrated EF improvements, whereas adults may require more specialized approaches that enable the flexible use of EF strategic or creative decision making. Additionally, understanding the age-related changes in the structure and function of EFs is critical for interpreting research findings and designing effective EF interventions in sports and exercise. EF development has been argued to continue into early adulthood, and the effects of physical exercise on EF might vary depending on the individual's developmental stage. More specifically, age-related differences in EF are likely to influence how well athletes learn and retain motor skills, especially in cognitively demanding sports (e.g. soccer, basketball) or are able to follow-rules in these games. Hence, tailoring sport-practice, play interventions (Furley, 2021), and exercise plans to the described developmental stages will also likely be a fruitful endeavour to enhance skill acquisition and learning in sports.

Building on this, future research should further explore how specific types of EF training interact with developmental phases across childhood, adolescence, and early adulthood. Longitudinal studies that track EF changes across these stages in athletic populations are particularly needed to identify optimal time windows for targeted interventions. Moreover, research should investigate how the timing, content, and intensity of physical activity or sport-specific training can be adjusted to match the neurocognitive needs of each developmental stage. Understanding these dynamics can support the creation of age-sensitive coaching practices and training programmes that maximize both cognitive and athletic development.

## Recommendation 10: Enhance Participant Diversity

> Addressing demographic diversity in EF research is essential to improve the generalizability of findings and uncover population-specific effects. Therefore, research should integrate balanced and representative samples overall, capturing a wide range of demographic factors, and explore the interplay of cultural, socioeconomic, educational, sex and gender factors in shaping EFs.

Most EF research has been conducted with limited demographic diversity, focusing on homogeneous groups. This lack of diversity limits the generalizability of the findings and hampers the identification of population-specific effects (Furley et al., 2023).

Socioeconomic status (SES) influences cognitive development and performance and thus also EFs (Hackman et al., 2015). Low SES is associated with challenges that negatively impact EF development, as children with lower SES utilize EF processes less effectively than their higher-SES peers (Noble et al., 2005). These disparities extend from childhood into adulthood, with some effects being irreversible (Hackman et al., 2010). Family income is a key predictor of cognitive performance, with prolonged economic hardship, particularly in rural areas, having detrimental effects (Korenmann et al., 1995; McLoyd, 1998). Children from lower SES backgrounds often exhibit working memory deficits, which impair overall EF development (Noble et al., 2005). Chronic poverty also increases allostatic load, negatively affecting the prefrontal cortex and disrupting cognitive functions (Lupien et al., 2007, 2015).

Educational experiences also play a crucial role in shaping EFs. While improvements in EFs are observed through specific training programmes, the transfer of these gains to broader contexts remains limited (Diamond & Lee, 2011). EFs correlate with academic performance and are more pronounced at higher education levels (Purpura et al., 2017). Cultural differences impact EFs, with Western cultures favouring analytical and categorical thinking, while Eastern cultures adopt a holistic and covariation-based approach. These differences stem from distinct social systems and problem-solving environments (Vygotsky, 1978). To enhance our understanding of these variations, future research should examine how cultural frameworks influence the development of EFs across diverse populations. Moreover, adopting hybrid approaches that integrate the strengths of both cultural paradigms could promote EF development globally.

Research on gender and sex differences in EFs shows limited evidence for significant sex-based discrepancies, with individual developmental trajectories playing a more substantial role in shaping EF outcomes (Grissom & Reyes, 2019). However, the sex data gap in research has hindered a comprehensive understanding of how sex differences may influence EF, especially in relation to athletic performance (Cowley et al., 2021). Future research should prioritize the inclusion of balanced and representative samples overall, considering biological sex to assess potential sex differences in EFs, particularly in relation to athletic performance, and gender as a broader social factor to capture diversity across populations. Importantly, these findings highlight that future research should also systematically investigate underlying neural mechanisms, as males and females may engage different brain networks or cognitive strategies during EF tasks – understanding these differences is crucial for developing more precise and individualized approaches in both research and applied sports settings (Gaillard et al., 2021).

Addressing demographic diversity in EF research is essential to improve the generalizability of findings and uncover population-specific effects. Future studies should integrate diverse participant samples, investigate interventions tailored to specific demographic groups, and explore the interplay of cultural, socioeconomic, educational, and gender factors in shaping EFs.

## Conclusion

The study of EFs in sport and exercise is a dynamic and multifaceted field that requires careful and methodical investigation. The ten recommendations outlined in this paper aim to provide a framework for enhancing the quality and precision of research in this area (see Figure 1 for a concise summary of all recommendations). These recommendations move from methodological foundations to contextual factors and population-level considerations, thereby providing both immediate guidance for improving current research and priorities for shaping the future of the field. However, it is important to note that this list is not exhaustive and does not represent a gold-standard approach to follow. Instead, we hope that these recommendations will serve as a guide for EF researchers in sport and exercise helping them to formulate more focused research questions, design better-controlled experiments, and generate more robust research findings. It is also hoped that these recommendations support practitioners assessing EF within sporting organisations. To do this, practitioners adhere to a set of minimum standards for EF assessment (e.g. construct-task alignment, basic reliability, and avoid over-interpreting single-task outcomes) and avoid assessing or training EF when the task lacks validity or the results are likely to be misinterpreted.


**Figure 1.** Overview of the ten recommendations for improving executive function research in sport and exercise.

By establishing clear, unified definitions of EFs, employing standardized measurement tools, and using ecologically valid tasks, researchers can ensure greater consistency and real-world relevance in their studies. In addition, incorporating longitudinal designs and fostering multidisciplinary collaboration will strengthen study design and accelerate cumulative progress. Moreover, controlling for variables such as fitness levels and physical fatigue, and distinguishing between different exercise modalities, will deepen our understanding of the long-term effects of exercise on cognitive performance and better inform causal inferences.

Attention to factors such as age, developmental stages, consideration of psychological moderators and mediators, and enhancement of participant diversity will allow for more tailored and insightful interventions. Promoting interdisciplinary collaboration and adopting these combined methodological and strategic priorities will further strengthen research by ensuring inclusivity and a more holistic approach. These recommendations are intended to encourage thoughtful and comprehensive research practices that will ultimately contribute to the development of evidence-based, personalized interventions aimed at optimizing cognitive and physical performance across diverse populations. While these recommendations are not definitive, we believe they will guide researchers toward generating more meaningful and reliable insights into the relationship between sport, exercise, physical activity and EFs. Future progress will also depend on transparent reporting, replication across diverse contexts, and stronger links between laboratory methods and applied sport settings, ensuring that EF research can meaningfully inform both theory and practice.

## Note

1. Drift-diffusion parameters are model-based indices derived from reaction time and accuracy data, reflecting the speed and quality of information accumulation in decision-making tasks, and often show higher reliability than raw difference scores.

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## Data Availability Statement

No new data were created or analysed for this article. Data sharing is not applicable to this article.

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## References

Alloway, T. P., Gathercole, S. E., Kirkwood, H., & Elliott, J. (2008). Evaluating the validity of the automated working memory assessment. *Educational Psychology*, *28*(7), 725–734. https://doi.org/10.1080/01443410802243828

Anderson, P. (2002). Assessment and development of executive function (EF) during childhood. *Child Neuropsychology*, *8*(2), 71–82. doi:10.1076/chin.8.2.71.8724

Anguera, J. A., Bernard, J. A., Jaeggi, S. M., Buschkuehl, M., Benson, B. L., Jennett, S., & Seidler, R. D. (2012). The effects of working memory resource depletion and training on sensorimotor adaptation. *Behavioural Brain Research*, *228*(1), 107–115. doi:10.1016/j.bbr.2011.11.040

Anguera, J. A., Boccanfuso, J., Rintoul, J. L., Al-Hashimi, O., Faraji, F., Janowich, J., & … Gazzaley, A. (2013). Video game training enhances cognitive control in older adults. *Nature*, *501*(7465), 97–101. https://doi.org/10.1038/nature12486

Araujo, D., Davids, K., & Passos, P. (2007). Ecological validity, representative design, and correspondence between experimental task constraints and behavioral setting: Comment on. *Ecological Psychology*, *19*(1), 69–78. doi:10.1080/10407410709336951

Baddeley, A. D., & Hitch, G. J. (1974). Working memory. In G. H. Bower (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 8, pp. 47–89). Academic Press.

Bailey, R. P., Madigan, D. J., Cope, E., & Nicholls, A. R. (2018). The prevalence of pseudoscientific ideas and neuromyths among sports coaches. *Frontiers in Psychology*, *9*, 641. https://doi.org/10.3389/fpsyg.2018.00641

Bangsbo, J., Iaia, F. M., & Krustrup, P. (2008). The Yo-Yo Intermittent Recovery Test: A useful tool for evaluation of physical performance in intermittent sports. *Sports Medicine*, *38*(1), 37–51. https://doi.org/10.2165/00007256-200838010-00004

Best, J. R., & Miller, P. H. (2010). A developmental perspective on executive function. *Child Development*, *81*(6), 1641–1660. doi:10.1111/j.1467-8624.2010.01499.x

Biagianti, B., & Vinogradov, S. (2013). Computerized cognitive training targeting brain plasticity in schizophrenia. *Progress in Brain Research*, *207*, 301–326. https://doi.org/10.1016/B978-0-444-63327-9.00011-4

Boot, W. R., Simons, D. J., Stothart, C., & Stutts, C. (2013). The pervasive problem with placebos in psychology: Why active control groups are not sufficient to rule out placebo effects. *Perspectives on Psychological Science*, *8*(4), 445–454. https://doi.org/10.1177/1745691613491271

Borg, G. A. (1982). Psychophysical bases of perceived exertion. *Medicine and Science in Sports and Exercise*, *14*(5), 377–381. https://doi.org/10.1249/00005768-198205000-00012

Brimmell, J., Edwards, E. J., Smith, M., & Vaughan, R. S. (2021). Think, see, do: Executive function, visual attention, and soccer penalty performance. *Sport, Exercise, and Performance Psychology*, *10*(2), 290. doi:10.1037/spy0000265

Brimmell, J., Edwards, E. J., Uiga, L., Wood, G., & Vaughan, R. S. (2025). The structure of executive functions in athletes: A latent variable approach. *Psychology of Sport and Exercise*, *78*, 102808. https://doi.org/10.1016/j.psychsport.2025.102808

Brimmell, J., Edwards, E. J., & Vaughan, R. S. (2022). Executive function and visual attention in sport: A systematic review. *International Review of Sport and Exercise Psychology*, 1–34. doi:10.1080/1750984X.2022.2145574

Brisswalter, J., Collardeau, M., & René, A. (2002). Effects of acute physical exercise characteristics on cognitive performance. *Sports Medicine (Auckland, N.Z.)*, *32*(9), 555–566. https://doi.org/10.2165/00007256-200232090-00002

Bürki, C. N., Ludwig, C., Chicherio, C., & de Ribaupierre, A. (2014). Individual differences in cognitive plasticity: An investigation of training curves in younger and older adults. *Psychological Research*, *78*(6), 821–835. doi:10.1007/s00426-014-0559-3

Casey, B. J., Tottenham, N., Liston, C., & Durston, S. (2005). Imaging the developing brain: What have we learned about cognitive development? *Trends in Cognitive Sciences*, *9*(3), 104–110. doi:10.1016/j.tics.2005.01.011

Chein, J. M., & Morrison, A. B. (2010). Expanding the mind's workspace: Training and transfer effects with a complex working memory span task. *Psychonomic Bulletin & Review*, *17*(2), 193–199. doi:10.3758/PBR.17.2.193

Collins, A., & Koechlin, E. (2012). Reasoning, learning, and creativity: Frontal lobe function and human decision-making. *PLoS Biology*, *10*(3), e1001293. https://doi.org/10.1371/journal.pbio.1001293

Cowley, E. S., Olenick, A. A., McNulty, K. L., & Ross, E. Z. (2021). "Invisible sportswomen": The sex data gap in sport and exercise science research. *Women in Sport and Physical Activity Journal*, *29*(2), 146–151. https://doi.org/10.1123/wspaj.2021-0028

Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, *64*(1), 135–168. doi:10.1146/annurev-psych-113011-143750

Diamond, A. (2016). Why improving and assessing executive functions early in life is critical. In J. A. Griffin, P. McCardle, & L. S. Freund (Eds.), *Executive function in preschool-age children: Integrating measurement, neurodevelopment, and translational research* (pp. 11–43). American Psychological Association.

Diamond, A. (2020). Executive functions. In J.L. Michaud, C. Bulteau, D. Cohen, & A. Gallagher (Eds.), *Handbook of clinical neurology* (Vol. 173, pp. 225–240). Elsevier.

Diamond, A., & Lee, K. (2011). Interventions shown to aid executive function development in children 4 to 12 years old. *Science*, *333*(6045), 959–964. https://doi.org/10.1126/science.1204529

Donnan, K., Williams, E. L., Morris, J. L., & Stanger, N. (2021). The effects of exercise at different temperatures on cognitive function: A systematic review. *Psychology of Sport and Exercise*, *54*, 101908. doi:10.1016/j.psychsport.2021.101908

Dwan, K., Altman, D. G., Arnaiz, J. A., Bloom, J., Chan, A. W., Cronin, E., Decullier, E., Easterbrook, P. J., Von Elm, E., & Gamble, C. (2008). Systematic review of the empirical evidence of study publication bias and outcome reporting bias. *PLoS One*, *3*(8), e3081. https://doi.org/10.1371/journal.pone.0003081

Eslinger, P. J. (1996). Conceptualizing, describing, and measuring components of executive function: A summary. In G. R. Lyon, & N. A. Krasnegor (Eds.), *Attention, memory and executive function* (pp. 367–395). Brookes.

Finkenzeller, T., Würth, S., Heilmann, F., Lautenbach, F., & Amesberger, G. (2023). Editorial: Executive functions in sports. *Frontiers in Sports and Active Living*, *5*, 1340244. https://doi.org/10.3389/fspor.2023.1340244

Foster, J. L., Harrison, T. L., Hicks, K. L., Draheim, C., Redick, T. S., & Engle, R. W. (2017). Do the effects of working memory training depend on baseline ability level? *Journal of Experimental Psychology. Learning, Memory, and Cognition*, *43*(11), 1677–1689. https://doi.org/10.1037/xlm0000426

Franceschini, S., Gori, S., Ruffino, M., Viola, S., Molteni, M., & Facoetti, A. (2013). Action video games make dyslexic children read better. *Current Biology*, *23*(6), 462–466. https://doi.org/10.1016/j.cub.2013.01.044

Friedman, N. P., & Miyake, A. (2017). Unity and diversity of executive functions: Individual differences as a window on cognitive structure. *Cortex*, *86*, 186–204. https://doi.org/10.1016/j.cortex.2016.04.023

Friedman, N. P., & Robbins, T. W. (2022). The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology*, *47*(1), 72–89. doi:10.1038/s41386-021-01132-0

Furley, P. (2021). The nature and culture of physical activity play: Towards a biocultural model of play, practice, and teaching in sports. *Evolutionary Behavioral Sciences*, *15*(2), 208. doi:10.1037/ebs0000254

Furley, P., Schul, K., & Memmert, D. (2016). Das Experten-Novizen-Paradigma und die Vertrauenskrise in der Psychologie. *Zeitschrift für Sportpsychologie*, *23*(4), 131–140. https://doi.org/10.1026/1612-5010/a000174

Furley, P., Schütz, L. M., & Wood, G. (2023). A critical review of research on executive functions in sport and exercise. *International Review of Sport and Exercise Psychology*, 1–29. https://doi.org/10.1080/1750984X.2023.2217437

Gaillard, A., Fehring, D. J., & Rossell, S. L. (2021). Sex differences in executive control: A systematic review of functional neuroimaging studies. *The European Journal of Neuroscience*, *53*(8), 2592–2611. https://doi.org/10.1111/ejn.15107

Gigerenzer, G. (2017). A theory integration program. *Decision*, 4(3), 133–145. https://doi.org/10.1037/dec0000082

Green, C. S., Bavelier, D., Kramer, A. F., Vinogradov, S., Ansorge, U., Ball, K. K., … Witt, C. M. (2019). Improving methodological standards in behavioral interventions for cognitive enhancement. *Journal of Cognitive Enhancement*, *3*(1), 2–29. doi:10.1007/s41465-018-0115-y

Grissom, N. M., & Reyes, T. M. (2019). Let's call the whole thing off: Evaluating gender and sex differences in executive function. *Neuropsychopharmacol*, *44*(1), 86–96. https://doi.org/10.1038/s41386-018-0179-5

Guye, S., De Simoni, C., & von Bastian, C. C. (2017). Do individual differences predict change in cognitive training performance? A latent growth curve modeling approach. *Journal of Cognitive Enhancement*, *1*(4), 374–393. doi:10.1007/s41465-017-0049-9

Hackman, D. A., Farah, M. J., & Meaney, M. J. (2010). Socioeconomic status and the brain: Mechanistic insights from human and animal research. *Nature Reviews. Neuroscience*, *11*(9), 651–659. https://doi.org/10.1038/nrn2897

Hackman, D. A., Gallop, R., Evans, G. W., & Farah, M. J. (2015). Socioeconomic status and executive function: Developmental trajectories and mediation. *Developmental Science*, *18*(5), 686–702. https://doi.org/10.1111/desc.12246

Hallock, H., Collins, D., Lampit, A., Deol, K., Fleming, J., & Valenzuela, M. (2016). Cognitive training for post-acute traumatic brain injury: A systematic review and meta-analysis. *Frontiers in Human Neuroscience*, *10*, 537. https://doi.org/10.3389/fnhum.2016.00537

Harris, D. J., Wilson, M. R., & Vine, S. J. (2018). A systematic review of commercial cognitive training devices: Implications for use in sport. *Frontiers in Psychology*, *9*, 312023. https://doi.org/10.3389/fpsyg.2018.00709

Herold, F., Törpel, A., Schega, L., & Müller, N. G. (2019). Functional and/or structural brain changes in response to resistance exercises and resistance training lead to cognitive improvements–A systematic review. *European Review of Aging and Physical Activity*, *16*(1), 1–33. https://doi.org/10.1186/s11556-019-0217-2

Huijgen, B. C., Leemhuis, S., Kok, N. M., Verburgh, L., Oosterlaan, J., Elferink-Gemser, M. T., & Visscher, C. (2015). Cognitive functions in elite and sub-elite youth soccer players aged 13 to 17 years. *PLoS One*, *10*(12), e0144580. https://doi.org/10.1371/journal.pone.0144580

Jaeggi, S. M., Buschkuehl, M., Jonides, J., & Perrig, W. J. (2008). Improving fluid intelligence with training on working memory. *Proceedings of the National Academy of Sciences*, *105*(19), 6829–6833. https://doi.org/10.1073/pnas.0801268105

Jaeggi, S. M., Buschkuehl, M., Jonides, J., & Shah, P. (2011). Short- and long-term benefits of cognitive training. *Proceedings of the National Academy of Sciences of the United States of America*, *108*(25), 10081–10086. doi:10.1073/pnas.1103228108

Kalén, A., Bisagno, E., Musculus, L., Raab, M., Pérez-Ferreirós, A., Williams, A. M., Araújo, D., Lindwall, M., & Ivarsson, A. (2021). The role of domain-specific and domain-general cognitive functions and skills in sports performance: A meta-analysis. *Psychological Bulletin*, *147*(12), 1290–1308. doi:10.1037/bul0000355

Karr, J. E., Areshenkoff, C. N., Rast, P., Hofer, S. M., Iverson, G. L., & Garcia-Barrera, M. A. (2018). The unity and diversity of executive functions: A systematic review and re-analysis of latent variable studies. *Psychological Bulletin*, *144*(11), 1147–1185. https://doi.org/10.1037/bul0000160

Katz, B., Shah, P., & Meyer, D. E. (2018). How to play 20 questions with nature and lose: Reflections on 100 years of brain-training research. *Proceedings of the National Academy of Sciences*, *115*(40), 9897–9904. doi:10.1073/pnas.1617102114

Klingberg, T., Fernell, E., Olesen, P. J., Johnson, M., Gustafsson, P., Dahlstrom, K., Gillberg, C., Forssberg, H., & Westerberg, H. (2005). Computerized training of working memory in children with ADHD—a randomized, controlled trial. *Journal of the American Academy of Child and Adolescent Psychiatry*, *44*(2), 177–186. doi:10.1097/00004583-200502000-00010

Korenmann, S., Miller, J., & Sjaastad, J. (1995). Long-term poverty and child development: Evidence from the NLSY. *Children and Youth Services Review*, *17*(1-2), 127–155. doi:10.1016/0190-7409(95)00006-X

Kovács, K., & Conway, A. R. (2016). Process overlap theory: A critical review and new synthesis of cognitive terminology and intelligence. *Psychological Inquiry*, *27*(3), 151–177. https://doi.org/10.1080/1047840X.2016.1153946

Lambourne, K., & Tomporowski, P. (2010). The effect of exercise-induced arousal on cognitive task performance: A meta-regression analysis. *Brain Research*, *1341*, 12–24. https://doi.org/10.1016/j.brainres.2010.03.091

Lezak, M. D. (1982). The problem of assessing executive functions. *International Journal of Psychology*, *17*(1-4), 281–297. doi:10.1080/00207598208247445

Liston, C., McEwen, B. S., & Casey, B. J. (2009). Psychosocial stress reversibly disrupts prefrontal processing and attentional control. *Proceedings of the National Academy of Sciences*, *106*(3), 912–917. doi:10.1073/pnas.0807041106

Liu, S., Yu, Q., Li, Z., Cunha, P. M., Zhang, Y., Kong, Z., Lin, W., Chen, S., & Cai, Y. (2020). Effects of acute and chronic exercises on executive function in children and adolescents: A systemic review and meta-analysis. *Frontiers in Psychology*, *11*, 554915. doi:10.3389/fpsyg.2020.554915

Luciana, M., & Nelson, C. A. (2002). Assessment of neuropsychological function through use of the Cambridge neuropsychological testing automated battery: Performance in 4- to 12-year-old children. *Developmental Neuropsychology*, *22*(3), 595–624. https://doi.org/10.1207/S15326942DN2203_3

Lupien, S. J., Maheu, F., Tu, M., Fiocco, A., & Schramek, T. E. (2007). The effects of stress and stress hormones on human cognition: Implications for the field of brain and cognition. *Brain and Cognition*, *65*(3), 209–237. doi:10.1016/j.bandc.2007.02.007

Lupien, S. J., Ouellet-Morin, I., Hupbach, A., Tu, M. T., Buss, C., Walker, D., Pruessner, J., & Mcewen, B. S. (2015). Beyond the stress concept: Allostatic load—A developmental biological and cognitive perspective. In D. Cicchetti & D. J. Cohen (Eds.), *Developmental psychopathology* (pp. 578–628). Wiley. doi:10.1002/9780470939390.ch14

Mashburn, C. A., Barnett, M. K., & Engle, R. W. (2024). Processing speed and executive attention as causes of intelligence. *Psychological Review*, *131*(3), 664–694. doi:10.1037/rev0000439

Mashburn, C. A., Tsukahara, J. S., & Engle, R. W. (2020). Individual differences in attention control. *Working Memory*, 175–211. https://doi.org/10.1093/oso/9780198842286.003.0007

McLoyd, V. C. (1998). Socioeconomic disadvantage and child development. *American psychologist*, *53*(2), 185. doi:10.1037/0003-066X.53.2.185

McMorris, T., Turner, A., Hale, B. J., & Sproule, J. (2016). Beyond the catecholamines hypothesis for an acute exercise–cognition interaction: A neurochemical perspective. In T. McMorris (Ed.), *Exercise-cognition interaction: Neuroscience perspectives* (pp. 65–103). Elsevier Academic Press.

Melby-Lervåg, M., & Hulme, C. (2013). Is working memory training effective? A meta-analytic review. *Developmental Psychology*, *49*(2), 270–291. https://doi.org/10.1037/a0028228

Mischel, W., Shoda, Y., & Rodriguez, M. L. (1989). Delay of gratification in children. *Science*, *244*(4907), 933–938. doi:10.1126/science.2658056

Miyake, A., & Friedman, N. P. (2012). The nature and organization of individual differences in executive functions: Four general conclusions. *Current Directions in Psychological Science*, *21*(1), 8–14. https://doi.org/10.1177/0963721411429458

Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive Psychology*, *41*(1), 49–100. https://doi.org/10.1006/cogp.1999.0734

Miyake, A., & Shah, P. (1999). *Models of working memory: Mechanisms of active maintenance and executive control*. Cambridge University Press.

Moreau, D. (2022). How malleable are cognitive abilities? A critical perspective on popular brief interventions. *American Psychologist*, *77*(3), 409–423. https://doi.org/10.1037/amp0000872

Nilsson, J. P., Söderström, M., Karlsson, A. U., Lekander, M., Åkerstedt, T., Lindroth, N. E., & Axelsson, J. (2005). Less effective executive functioning after one night's sleep deprivation. *Journal of Sleep Research*, *14*(1), 1–6. doi:10.1111/j.1365-2869.2005.00442.x

Noble, K. G., Norman, M. F., & Farah, M. J. (2005). Neurocognitive correlates of socioeconomic status in kindergarten children. *Developmental Science*, *8*(1), 74–87. https://doi.org/10.1111/j.1467-7687.2005.00394.x

Nyongesa, M. K., Ssewanyana, D., Mutua, A. M., Chongwo, E., Scerif, G., Newton, C. R., & Abubakar, A. (2019). Assessing executive function in adolescence: A scoping review of existing measures and their psychometric robustness. *Frontiers in Psychology*, *10*, 311. https://doi.org/10.3389/fpsyg.2019.00311

O'Hare, E. D., & Sowell, E. R. (2008). Imaging developmental changes in gray and white matter in the human brain. In C. A. Nelson & M. Luciana (Eds.), *Handbook of developmental cognitive neuroscience* (2nd ed., pp. 23–38). MIT Press.

Packwood, S., Hodgetts, H. M., & Tremblay, S. (2011). A multiperspective approach to the conceptualization of executive functions. *Journal of Clinical and Experimental Neuropsychology*, *33*(4), 456–470. https://doi.org/10.1080/13803395.2010.533157

Panesi, S., Bandettini, A., Traverso, L., & Morra, S. (2022). On the relation between the development of working memory updating and working memory capacity in preschoolers. *Journal of Intelligence*, *10*(1), 5. https://doi.org/10.3390/jintelligence10010005

Parry, T. E., Myszka, S., Yearby, T., O Sullivan, M., & Otte, F. (2025). The value of opposed and unopposed practice: An ecological dynamics rationale for skill development. *Quest (Grand Rapids, Mich )*, 1–21. doi:10.1080/00336297.2024.2420759

Pribram, K. H. (1973). The primate frontal cortex–executive of the brain. In K. H. Pribram & A. R. Luria (Eds.), *Psychophysiology of the frontal lobes* (pp. 293–314). Academic Press.

Purpura, D. J., Schmitt, S. A., & Ganley, C. M. (2017). Foundations of mathematics and literacy: The role of executive functioning components. *Journal of experimental child psychology*, *153*, 15–34. https://doi.org/10.1016/j.jecp.2016.08.010

Roca, A., & Ford, P. R. (2020). Decision-making practice during coaching sessions in elite youth football across European countries. *Science and Medicine in Football*, *4*(4), 263–268. doi:10.1080/24733938.2020.1755051

Rohde, T. E., & Thompson, L. A. (2007). Predicting academic achievement with cognitive ability. *Intelligence*, *35*(1), 83–92. doi:10.1016/j.intell.2006.05.004

Sala, G., & Gobet, F. (2019). Cognitive training does not enhance general cognition. *Trends in Cognitive Sciences*, *23*(1), 9–20. https://doi.org/10.1016/j.tics.2018.10.004

Sambol, S., Suleyman, E., Scarfo, J., & Ball, M. (2023). A true reflection of executive functioning or a representation of task-specific variance? Re-evaluating the unity/diversity framework. *Acta psychologica*, *236*, 103934. https://doi.org/10.1016/j.actpsy.2023.103934

Serpell, Z. N., & Esposito, A. G. (2016). Development of executive functions: Implications for educational policy and practice. *Policy Insights from the Behavioral and Brain Sciences*, *3*(2), 203–210. https://doi.org/10.1177/2372732216654718

Shah, P., Buschkuehl, M., Jaeggi, S. M., & Jonides, J. (2012). Cognitive training for ADHD: The importance of individual differences. *Journal of Applied Research in Memory and Cognition*, *1*(3), 204–205. doi:10.1016/j.jarmac.2012.07.001

Shipstead, Z., Redick, T. S., & Engle, R. W. (2012). Is working memory training effective? *Psychological Bulletin*, *138*(4), 628–654. https://doi.org/10.1037/a0027473

Simons, D. J., Boot, W. R., Charness, N., Gathercole, S. E., Chabris, C. F., Hambrick, D. Z., & Stine-Morrow, E. A. (2016). Do “brain-training” programs work? *Psychological Science in the Public Interest: a Journal of the American Psychological Society*, *17*(3), 103–186. https://doi.org/10.1177/1529100616661983

Simpson, A., Riggs, K. J., Beck, S. R., Gorniak, S. L., Wu, Y., Abbott, D., & Diamond, A. (2012). Refining the understanding of inhibitory processes: How response prepotency is created and overcome. *Developmental Science*, *15*(1), 62–73. https://doi.org/10.1111/j.1467-7687.2011.01105.x

Stieff, M., & Uttal, D. (2015). How much can spatial training improve STEM achievement. *Educational Psychology Review*, *27*(4), 607–615. doi:10.1007/s10648-015-9304-8

Stillman, C. M., Cohen, J., Lehman, M. E., & Erickson, K. I. (2016). Mediators of physical activity on neurocognitive function: A review at multiple levels of analysis. *Frontiers in Human Neuroscience*, *10*, 626. https://doi.org/10.3389/fnhum.2016.00626

Studer-Luethi, B., Jaeggi, S. M., Buschkuehl, M., & Perrig, W. J. (2012). Influence of neuroticism and conscientiousness on working memory training outcome. *Personality and Individual Differences*, *53*(1), 44–49. doi:10.1016/j.paid.2012.02.012

Verbruggen, F., Aron, A. R., Band, G. P., Beste, C., Bissett, P. G., Brockett, A. T., Brown, J. W., Chamberlain, S. R., Chambers, C. D., Colonius, H., Colzato, L. S., Corneil, B. D., Coxon, J. P., Dupuis, A., Eagle, D. M., Garavan, H., Greenhouse, I., Heathcote, A., Huster, R. J., … Boehler, C. N. (2019). A consensus guide to capturing the ability to inhibit actions and impulsive behaviors in the stop-signal task. *eLife*, *8*, e46323. https://doi.org/10.7554/eLife.46323

Vestberg, T., Reinebo, G., Maurex, L., Ingvar, M., & Petrovic, P. (2017). Core executive functions are associated with success in young elite soccer players. *PLoS One*, *12*(2), e0170845. https://doi.org/10.1371/journal.pone.0170845

Voss, M. W., Kramer, A. F., Basak, C., Prakash, R. S., & Roberts, B. (2010). Are expert athletes ‘expert’in the cognitive laboratory? A meta-analytic review of cognition and sport expertise. *Applied Cognitive Psychology*, *24*(6), 812–826. doi:10.1002/acp.1588

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (Vol. 86). Harvard University Press.

Watson, A. J., & Bell, M. A. (2013). Individual differences in inhibitory control skills at three years of age. *Developmental Neuropsychology*, *38*(1), 1–21. https://doi.org/10.1080/87565641.2012.718818

Waxer, M., & Morton, J. B. (2011). The development of future-oriented control: An electrophysiological investigation. *NeuroImage*, *56*(3), 1648–1654. doi:10.1016/j.neuroimage.2011.02.001

Westfall, J., & Yarkoni, T. (2016). Statistically controlling for confounding constructs is harder than you think. *PLoS One*, *11*(3), e0152719. https://doi.org/10.1371/journal.pone.0152719

Whitlock, L. A., McLaughlin, A. C., & Allaire, J. C. (2012). Individual differences in response to cognitive training: Using a multi-modal, attentionally demanding game-based intervention for older adults. *Computers in Human Behavior*, *28*(4), 1091–1096. doi:10.1016/j.chb.2012.01.012

Wilke, J., Giesche, F., Klier, K., Vogt, L., Herrmann, E., & Banzer, W. (2019). Acute effects of resistance exercise on cognitive function in healthy adults: A systematic review with multilevel meta-analysis. *Sports medicine (Auckland, N.Z.)*, *49*(6), 905–916. https://doi.org/10.1007/s40279-019-01085-x

Wood, G., Schütz, L.-M., Uiga, L., Brimmell, J., & Furley, P. (2023). The BASES Expert Statement on the relationship between executive functions and sport and exercise. *The Sport and Exercise Scientist*, *78*, 24–27.

Wright, R., Thompson, W. L., Ganis, G., Newcombe, N. S., & Kosslyn, S. M. (2008). Training generalized spatial skills. *Psychonomic Bulletin and Review*, *15*(4), 763–771. https://doi.org/10.3758/PBR.15.4.763

Yangüez, M., Bediou, B., Chanal, J., & Bavelier, D. (2023). In search of better practice in executive functions assessment: Methodological issues and potential solutions. *Psychological Review*, *131*(2), 402–430. doi:10.1037/rev0000434

Yu, Q., Chan, C. C. H., Chau, B., & Fu, A. S. N. (2017). Motor skill experience modulates executive control for task switching. *Acta Psychologica*, *180*, 88–97. https://doi.org/10.1016/j.actpsy.2017.08.013

Zelazo, P. D., & Carlson, S. M. (2012). Hot and cool executive function in childhood and adolescence: Development and plasticity. *Child Development Perspectives*, *6*(4), 354–360. doi:10.1111/j.1750-8606.2012.00246.x

Zhang, J., Tam, W. W. S., Hounsri, K., Kusuyama, J., & Wu, V. X. (2024). Effectiveness of combined aerobic and resistance exercise on cognition, metabolic health, physical function, and health-related quality of life in middle-aged and older adults with type 2 diabetes mellitus: A systematic review and meta-analysis. *Archives of Physical Medicine and Rehabilitation*, *105*(8), 1585–1599. https://doi.org/10.1016/j.apmr.2023.10.005

Zhou, B., Wang, Z., Zhu, L., Huang, G., Li, B., Chen, C., Huang, J., Ma, F., & Liu, T. C. (2022). Effects of different physical activities on brain-derived neurotrophic factor: A systematic review and Bayesian network meta-analysis. *Frontiers in Aging Neuroscience*, *14*, 981002. doi:10.3389/fnagi.2022.981002

Zinke, K., Zeintl, M., Eschen, A., Herzog, C., & Kliegel, M. (2011). Potentials and limits of plasticity induced by working memory training in old-old age. *Gerontology*, *58*(1), 79–87. https://doi.org/10.1159/000324240

Zinke, K., Zeintl, M., Rose, N. S., Putzmann, J., Pydde, A., & Kliegel, M. (2014). Working memory training and transfer in older adults: Effects of age, baseline performance, and training gains. *Developmental Psychology*, *50*(1), 304–315. https://doi.org/10.1037/a0032982